curriculum vitae

PERSONAL INFORMATION

Surname	**Agrawal**
Name	**Vivek**
Address	Maa Shitala Apartment, Baguiati, Kolkata, West Bengal, Pincode-700059, India
Telephone	+919007113430
E-mail	**vagrawal@kth.se**
Skype	**vivek8705**

Nationality	Indian

Date of birth	31st Dec, 1987

Education and training

• Date (From-To)	August 2010 – December 2012
• Name and type of organisation providing education and training	KTH, Royal Institute of Technology, SE - 100 44, Stockholm, Sweden.
Duration of the program of study	24 Months
• Principal subjects/occupational skills covered	Network Security, Vehicular Communication, Software Engineering and Security Architecture, Principle of Computer Security, Security in Mobile and Wireless Networks, Advanced Internetworking, Security Management, Network programming with java.
• Title of qualification awarded	Master of Science in Information and Communication Systems Security.
Final mark obtained	3.74 (Min - 2, Max - 4)

• Date (From-To)	August 2005 - August 2009
• Name and type of organisation providing education and training	Heritage Institute of Technology, West Bengal University of Technology, Chowbaga Road, Anandapur, Kolkata 700 107, India
Duration of the program of study	48 Months
• Principal subjects/occupational skills covered	Data Structure and Algorithms, Computer Organization and Architecture, Operating System, Formal Language and Automata, Software Engineering, Data Communication and Networking, Internetworking, Web Technology, Mobile Communication
• Title of qualification awarded	Bachelor of Technology in Information Technology
Final mark obtained	8.32 (Min - 5, Max - 10)

graduation thesis (Master)

Title	Performance evaluation of Group signature scheme – A feasibility study for Vehicular Communication
Language	English
Supervisor	Prof. Panos Papadimitratos
Thesis Summary	Users need to communicate with each other in many situations to share information. This creates the danger of the user's privacy being breached and it can discourage users from taking active participation in any information sharing task. In typical vehicular communication systems, users must remain anonymous and identity must not be revealed while exchanging safety and traffic related messages. Group signatures are versatile cryptographic tools that are suitable when we need security and privacy protection. A group signature scheme allows members of a group to sign messages on behalf of the group. Any receiver can verify the message validity but cannot discover the identity of the sender from the signed message or link two or more messages from the same signer. However, the identity of the signer can be discovered by an authority using a signed message. For this reason, Group Signature schemes were proposed in the context of vehicular communication systems. In this context, communication and computation overheads are critical. Thus, the focus of this thesis is to implement and compare different group signature schemes in terms of overhead introduced due to processing cost, and analytically evaluate their suitability for vehicular communication scenarios.

publications and articles submitted

Author(s) and title	Vivek Agrawal, "Performance evaluation of Group signature scheme – A feasibility study for Vehicular Communication"
Language	English
Publication place	Royal Institute of Technology, KTH
Date of publication	December 27th, 2012

Author(s) and title	Atul Chowdhary, Subhajit Karmakar, Vivek Agrawal, and Sandip Sarkar, "Investigation of Task based Gesture Recognition for Laser Actuated Presentation Systems"
Language	English
Publication place	6th IASTED International Conference on Human-Computer Interaction (IASTED HCI 2011), Washington, DC, USA
Date of publication	May 16 – 18, 2011

Author(s) and title	Atul Chowdhary, Vivek Agrawal, Subhajit Karmakar, and Sandip Sarkar, "Web Camera Based Laser Actuated Presentation System"
Language	English
Publication place	3rd International Conference on Human Computer Interaction (India HCI 2011), Bangalore, India
Date of publication	7-10 April, 2011

Work experience, stages, studies abroad

• Date (From-To)	January 2011 - Dec 2011
• Name and address of firm/university	Cinnober Financial Technology AB, Kungsgatan 36, SE-111 35 Stockholm, Sweden
• Type of business or sector	Financial Technology
• Type of employment	Test Engineer (Internship)
• Main activities and responsibilities	Automatic and Manual Testing, White Box and Black Box Testing using Java and Junit

• Date (From-To)	July 2011
• Name and address of firm/university	Ghent University, Jozef Plateaustraat 22, B-9000 Ghent, Belgium

• Type of business or sector		Summer Training school
• Type of employment		Training
• Main activities and responsibilities		Intellectual property protection of software, Software security via lightweight process virtualization, Theory and practice of code attack: Semantics, analysis and code transformation, Software security patterns, direct attack analysis and protection techniques, Software Protection Techniques against Hardware-Based Attacks on Software

Personal skills and competences

Mother tongue		**Hindi**

Other language(s)

		English
• reading		excellent
• writing		excellent
• speaking		excellent
		Swedish
• reading		Elementary
• writing		-
• speaking		Elementary

Social skills and competences		I have a good communication skill and I am skilled in public presentation including presenting technical data. I am able to establish and maintain good working relations with people of different national and cultural backgrounds. I have a good ability to adapt to multicultural environments, gained through my work experience in Sweden. I have worked with teams of different background, ethnicity and strength. I am able to live in worldwide locations.

Organisational skills and competences		I have a strong sense of leadership and I acted as a group leader for my teams in various technical events. I am highly organized and possess excellent planning skills to prioritize work. I worked as the Chairman (IEEE Member ID: 90409680) of IEEE Students' Branch, Heritage Chapter from March 2008 to March 2009. I have initiated various training programs and seminars for ROBOTICS

Technical skills and competences		Cryptography framework and terminologies Web security, TCP/IP, IPSEC, SSL, TLS, Kerberos, DES, RSA Risk Analysis and Management using OCTAVE-S method Security in Smart Phones OS (Android) Black Box and White Box Testing, Automatic and Manual Testing, Vulnerability assessment and Penetration Testing Detailed Knowledge of the common types of vulnerabilities such as the OWASP Top-Ten Network and wireless protocol (WIFI, WPA, EAPOL, Bluetooth, Ad Hoc networks, etc) Vehicular communication (802.11p), V2V, V2I Privacy enhancing technologies and Identity Management Java programming and JUnit test (Java Cryptography Architecture (JCA), Web application) using Netbeans and Eclipse Identifying software vulnerabilities using tools e.g. WebScarab, Fortify, etc. Search Engine Optimization concept and tools Network Traffic Analysis tools (Wireshark, etc.) Basic use of Traffic simulator (SUMO, OMNET++) Cisco Switch and Router

Technical Projects		Vulnerability in jForum Architecture; SQL Injection, XSS, CSRF, and other popular OWASP security risks. Information Security Risk Evaluation using OCTAVE S approach. Create and verify MAC and protect it using shared secret using JAVA. A survey on Privacy in Electronic Health Record Systems – Consumer's perspective. Developed web based 3 tiered Acme Web Shop that allowed a user to register, login, search for products, and add them in the shopping basket. A customer can check out products anytime from the shopping basket. We used Java, Primefaces3.0, EJB, JPA, JAVA DB (database) to develop this project. Designed a complete ISP with the help of routers, switches. We implemented dynamic IP routing (OSPF), fault tolerant IP routing (HSRP), IP multicast routing (PIM-SM). The internet application services which were implemented include DNS, DHCP, and Web server. We also used OpenVPN to implement VPN service in our ISP.

Artistic skills and competences		Digital photography, Mouthorgan, Guitar, Music, reading books, Drawing, cooking

Other skills and competences		Good coding skill in C, Java, HTML/CSS Good understanding of IBM SPSS, Eclipse, NetBeans, MATLAB Deep interest in Pattern Recognition, Image Segmentation, Human Computer Interaction, Interactive design Familiar with Windows, Linux, and Unix platforms.

Additional information		
Achievements		Recipient of Erasmus Mundus (Action II) Scholarship for Master's program at KTH ISSISP 2011 Half scholarship grant, Ghent University, Belgium I am among the top 3 students of my department during my Master's study. I am among the top 10 students during my undergraduate study. Secured 1st rank in Paper presentation in SPARX 09 held at ISM (Dhanbad, India). *TOPIC: Laser Actuated Mouse System*. Secured 2nd rank in SPARX 08 held at ISM (Dhanbad, India). *TOPIC: Write a program to design a traffic control system using 8085 and 8255 microprocessor.* Second runner up at IROBOTIX 07 held at HERITAGE INSTITUTE OF TECHNOLOGY (India). *TOPIC: Create manual control robot which can burst balloons present in the arena.* Third runner up in the inter college TECH TRIVIA event held at HERITAGE INSTITUTE OF TECHNOLOGY (India).
Personal Webpage		https://www.kth.se/profile/66265/ http://se.linkedin.com/pub/vivek-agrawal/31/331/b02/
References		Personal and professional references will be forwarded upon request.